

Professor, Korea University – Department of Electrical and Computer Engineering, Seoul, Korea

Visiting Professor, Seoul National University Hospital, Seoul, Korea

• E-mail: joongheon@korea.ac.kr • WWW: <https://joongheon.github.io>

Educational Backgrounds

- **University of Southern California (USC) – Viterbi School of Engineering** *Los Angeles, California, USA*
 - Ph.D. (08/2009–08/2014) in **Computer Science**, Thomas Lord Department of Computer Science
Advisor: Prof. Andreas F. Molisch (Communications, Information, Learning & Quantum (CLIQ) Group (Electrical Engineering))
 - M.S. (05/2014) in **Computer Science** with specialization in **High Performance Computing and Simulations**
 - M.S. (05/2012) in **Electrical Engineering**
 - **Korea University – College of Informatics** *Seoul, Republic of Korea*
 - M.S. (03/2004–02/2006) in **Computer Science and Engineering**
 - B.S. (03/1999–02/2004) in **Computer Science and Engineering**
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Professional Affiliations

Full-Time

- **Korea University**, *Seoul, Republic of Korea*
 - Professor (09/2019–Present)**, Department of Electrical and Computer Engineering
 - Adjunct Professor (03/2023–Present)**, Department of Communications Engineering (Samsung Electronics)
 - Adjunct Professor (03/2021–Present)**, Department of Semiconductor Engineering (SK Hynix)
 - Adjunct Professor (11/2022–Present)**, Department of Future Science and Technology Business (Graduate School)
 - Affiliated Professor (03/2025–Present)**, Department of Defense Convergence Technology (LIG Nex1, Graduate School)
- **Chung-Ang University**, *Seoul, Republic of Korea*
 - Assistant Professor (03/2016–08/2019)**, School of Computer Science and Engineering
- **Intel Corporation**, *Santa Clara (Silicon Valley), California, USA*
 - Systems Engineer (09/2013–02/2016)**, Platform Engineering Group
- **University of Southern California – Viterbi School of Engineering**, *Los Angeles, California, USA*
 - Ph.D. Student in Computer Science (08/2009–08/2014)**, Thomas Lord Department of Computer Science
 - Annenberg Graduate Fellow (2009)**, 4-Year Full Scholarship for the Ph.D. Program in Computer Science
 - Research Assistant (01/2011–08/2014)**, Communications, Information, Learning & Quantum (CLIQ) Group
 - Teaching Assistant (01/2012–05/2013)**, Computer Science and Electrical Engineering Major Courses
- **InterDigital**, *San Diego, California, USA*
 - Intern (05/2012–08/2012)**, Wireless Systems Evolution Department
- **LG Electronics – CTO Office**, *Seoul, Republic of Korea*
 - Research Engineer (01/2006–08/2009)**, Multimedia Research Lab, Seocho R&D Campus

Joint Appointments for Quantum Research Collaboration

- **Seoul National University Hospital**, *Seoul, Republic of Korea*
 - Visiting Professor (10/2025–Present)**, Healthcare AI Research Institute

Services for National R&D

- **Presidential Council on National Artificial Intelligence Strategy**, **Advisory Committee (12/2025–Present)**
 - **National Research Foundation of Korea**, **Review Board (11/2024–10/2026)**
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University R&D/Administrative Positions

R&D Leadership Positions

- **Korea University**, *Seoul, Republic of Korea*
 - Director (07/2024–12/2031)**, Net-Zero CAFE (Connectivity and Autonomy for Future Ecosystem) Research Center
 - Principal Investigator, Software Star-Lab (07/2024–12/2031)**, Quantum AI Empowered Second-Life Platform Technology

Administrative Positions

- **Korea University**, *Seoul, Republic of Korea*
 - Vice Department Chair (01/2025–08/2025)**, Academic Affairs, School of Electrical Engineering
 - Deputy Vice President (02/2022–08/2024)**, Office of Academic Affairs
 - Dean (06/2021–08/2023)**, Center for Teaching and Learning (CTL)

Committee Members

- **Korea University**, Seoul, Republic of Korea
Member (12/2023–Present), Council of the Interdisciplinary Program in Quantum Information Science
Member (03/2021–02/2027), Korea University 120th Anniversary Planning Committee
Member (06/2021–03/2022, 09/2022–08/2023), Academic Affair Planning Committee
Member (03/2021–08/2023), Distance Learning Management Committee
Steering Committee Member (07/2020–06/2022), Institute of Data Science (IDS)

Awards and Honors

Research and Academic Excellence (International)

- Top Cited Article in ETRI Journal (Wiley), among work published in 2024 2025
- Finalist, AAAI Student Abstract and Poster Session – Oral Presentation 2026 (Top 35), 2023 (Top 25)
- IEEE ICTC Best Paper Award – IEEE Communications Society 2022
- Spotlight, Oral Presentation – ICML Workshop on Dynamic Neural Networks (2022) 2022
- IEEE MMTC Best Journal Paper Award – IEEE Communications Society 2021
- IEEE ICOIN Best Paper Award – IEEE Computer Society 2021
- IEEE MMTC Outstanding Young Researcher Award – IEEE Communications Society 2020
- IEEE Systems Journal Best Paper Award – IEEE Systems Council 2020
- Next Generation and Standards (NGS) Division Recognition Award – Intel Corporation Q1/2015
- Annenberg Graduate Fellowship Award – University of Southern California 2009–2013
Awarded with Ph.D. Admission: 4 Year Full Scholarship (Tuition Waiver and \$120,000 Stipend (\$30,000/year for 4 years))
- IEEE Seoul Section Student Paper Contest Gold (2019), Silver (2025), Bronze (2025, 2024, 2024, 2023, 2020)
- IEEE VTS Seoul Chapter Award – IEEE Vehicular Technology Society 2023, 2022, 2022, 2021, 2021, 2019

Research and Academic Excellence (Korea Regional, since 2016)

- HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS 2025
- Best Paper Award, The Journal of KICS – KICS 2024
- HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS 2023
- Haedong Young Scholar Award – KICS and Haedong Foundation 2018
For recognizing a researcher under the age of 40 who has made outstanding contributions to IT R&D

Korea University

- Granite Tower Best Research Award (Top 3%), Totally, 2 times
– 2025 (Top 3%, top 53 professors at Korea University in 2025) 05/2026
– 2024 (Top 3%, top 54 professors at Korea University in 2024) 05/2025
- Best Research Achievement Award – Korea University, School of Electrical Engineering 2024
- Insung Research Grant Award – Korea University 01/2023
For recognizing professors in top 5% research excellence during the first 3 years at Korea University
- Granite Tower Best Teaching Award (Top 5%), Totally, 7 times
– Probability and Random Process KECE209-02 (2025)
– Software Programming Basics GECT002-01 (2024), GECT002-06 (2025)
– Computer Language and Laboratory EGRN151-07 (2019), EGRN151-06 (2021)
– Introduction to Computers SEMI103 (2021)
– Future Mobility Technology GEQR075 (2022)
- Best Teaching Award (Top 20%), Totally, 7 times
– Probability and Random Process KECE209-02 (2021), KECE209-02 (2022)
– Software Programming Basics GECT002-02 (2024), GECT002-08 (2024), GECT002-09 (2025)
– Computer Language and Laboratory EGRN151-09 (2020)
– Object-Oriented Programming SEMI104 (2021)

Academic and University Services (International)

- Certificate of Appreciation – IEEE Future Networks, AI/ML Working Group 08/2025
- Certificate of Appreciation – IEEE/IFIP WiOpt (2024) 10/2024
- Best Editor Award – ICT Express (Elsevier) 07/2023
- 2022 Best Chapter Award, IEEE Vehicular Technology Society Chapter, Awarded as a Treasure 2022
- Appreciation Recognition – Daegu Gyeongbuk Institute of Science and Technology (DGIST) 2021
- Certificate of Appreciation – Department of Computer Science, University of Southern California 2010

Academic and University Services (Korea Regional)

- Outstanding Contribution Award
– The Korean Institute of Communication Sciences and Information Sciences (KICS) 2025, 2024, 2021, 2019
– Korean Institute of Information Scientists and Engineers (KIISE) Information Network Society 2023, 2022
– Open Standards and ICT Association (OSIA) 2025, 2021
– The Institute of Electronics and Information Engineers (IEIE) 2025

- **Commemorative Plaque, KICS Excellent Lecture** (Title: Quantum Machine Learning and Software Technologies) 2025
- **Best Research Group Award, KICS Research Group in Quantum Communications and Quantum Computing** (Group Leader) 2025
- **Appreciation Recognition – Daegu Gyeongbuk Institute of Science and Technology (DGIST)** 2021

Student Awards (International)

- **Mitacs Globalink Research Award - for research in Canada**, Awarded to *Seok Bin Son* 2026
Project Title) Quantum Federated Learning in Distributed Systems: Challenges and Future Directions
Advisor) Prof. Trung Q. Duong (Memorial University of Newfoundland)
- **Mitacs Globalink Research Award - for research in Canada**, Awarded to *Seungcheol Oh* 2026
Project Title) Over the Air Quantum Federated Learning
Advisor) Prof. Octavia Dobre (Memorial University of Newfoundland)
- **IEEE TCSC Award for Excellence in Scalable Computing** (IEEE Computer Society), Awarded to *Soohyun Park* 2025
- **Young Scientist Award** (ICSANE 2025), Awarded to *Minji Lee* 2025
- **IEEE Vehicular Technology Society Student Scholarship Award (2025)**, Awarded to *Gyu Seon Kim* 2025
- **IEEE ICDCS Best Runner-Up Poster Paper Award (2025)**, Awarded to *Seok Bin Son, Soohyun Park* 2025

R&D Projects

University-Wide/Center/Flagship Projects

- **(Center Director, PI) Net-Zero CAFE Research Center** (07/2024–12/2031) ITRC (Korea Univ)
- **(SW Star-Lab, PI) Quantum AI Empowered Second-Life Platform Technology** (07/2024–12/2031) Star-Lab (Korea Univ)
- **Intelligent 6G Wireless Access System Research Center** (04/2021–12/2025) 6G AI Research Center (Korea Univ)
- **K-Starlink: Dynamic Reconfigurable and Intelligent Space-Terrestrial Networks** (06/2021–05/2024) Basic Research Lab (Ajou)
- **Nano UAV Intelligence Systems Research Lab** (10/2020–08/2023) ADD Military Special Research Center (Kwangwoon)
- **5G/Unmanned Vehicle Research Center (5G/UV-RC)** (06/2020–12/2022) ITRC (Hanyang)
- **Human Resource Development for the Biomedical Unstructured Big Data Analysis** (08/2018–12/2021) ITRC (SNU-Hospital)
- **Novel Data Science Driven Framework for Efficient Network Design** (06/2017–05/2020) Basic Research Lab (Chung-Ang)
- **Intelligent Internet of Energy (IoE) Data Research Center** (02/2020–05/2020) ITRC (Kookmin)

Consortium Projects

- **6GARROW: 6G AI-Native Integrated RAN-Core Networks** (09/2024–08/2027) IITP
- **AI Bots Collaborative Platform and Self-Organizing Artificial Intelligence Technology Development** (04/2022–12/2026) IITP
- **Development of Integrated Development Framework that supports Automatic Neural Network Generation and Deployment optimized for Runtime Environment** (04/2021–12/2025) IITP
- **Integrated Perception Technology Developments for Public Safety Platforms** (06/2019–05/2023) NRF
- **Development of Privacy-Reinforcing Distributed Transfer-Iterative Learning Algorithm** (07/2019–12/2021) MHW
- **Virtual Presence in Moving Objects through 5G (PriMO-5G)** (06/2018–06/2021) IITP
- **Distributed Secure Platform for Scalable Clinical OMOP CDM Models** (04/2019–12/2020) MHW

Individual Projects

- **Quantum-Empowered Spatio-Temporal Multi-Scale Digital Twin System** (03/2025–02/2028) NRF
- **Research on Trainability and Learnability in Quantum Machine Learning Models** (03/2026–10/2027) ETRI-Affiliated Institute
- **SEDNA: Satellite-aided Entanglement Distribution for Networked Applications** (10/2026–09/2027) NRF
- **Advancement Technology Development for Torpedo Deception Strategies in Submarines** (11/2022–11/2026) LIG Nex1
- **Advancement Tech Dev for Submarine Target Identification and Engagement Support Intelligence** (11/2022–11/2026) LIG Nex1
- **Research Planning for RL-based Scheduling in Autonomous Satellite Task Allocation** (05/2026–10/2026) KOITA
- **mmWave-based Liquid Classification in Next-Gen Robotic Vacuum Cleaners** (06/2026–09/2026) LG Electronics (Robotics Lab)
- **Mapping between Real World and VR for End-Edged Cloud Real-Time VR Servers** (09/2020–11/2025) Samsung Electronics
- **LEO Satellite Routing Research using Large Language Model and Reinforcement Learning** (05/2025–11/2025) ETRI
- **Discovering Quantum-Advantage Cases for Industry Adoption** (09/2025–11/2025) KTL
- **Research on Generative Quantum Machine Learning Models** (04/2025–10/2025) ETRI-Affiliated Institute
- **Quantum Hyper-Driving: Quantum-Inspired Hyper-Connected and Hyper-Sensing Autonomous Mobility** (03/2022–02/2025) NRF
- **Research on Learning-based Swarm Mission Planning Algorithms** (03/2024–02/2025) LIG Nex1
- **Quantum Reinforcement Learning for Satellite Backhaul Routing in Disaster Networks** (05/2024–11/2024) ETRI
- **Korea-Japan Joint Seminar Project for Generative and Multi-Modal AI Technologies** (10/2023–09/2024) NRF
- **Quantum Machine Learning-based Objection Detection for Point Cloud and its Acceleration** (12/2022–04/2024) Hyundai Motors
- **NOMA-based Resource Allocation Research in Space-Air-Ground Integrated Networks** (09/2023–11/2023) ETRI
- **Routing Algorithms for LEO Satellite Networks** (12/2022–08/2023) Solvit System
- **Optimal Positioning Algorithms for Wide-Area Relaying Networks** (12/2022–08/2023) Solvit System
- **Distributed Learning Algorithms to Build AI Models with Multi-Center Clinical Data** (11/2022–02/2023) Cipherome
- **Cellular/Wi-Fi Handover Technology Development** (02/2022–12/2022) LG Electronics CTO Division (Smart Mobility Lab)
- **Autonomous Intelligent COA Search Methods for Cyber-Attacks** (12/2021–11/2022) ADD
- **Development of Quantum Deep Reinforcement Learning Algorithm using QAOA** (10/2019–04/2022) NRF
- **Research Trends in Digital Twin Applications to Autonomous Driving** (03/2022–04/2022) Hyundai NGV
- **Distributed Learning System Design and Implementation for Clinical Applications** (02/2022–03/2022) Cipherome
- **mmWave Radar and Deep Reinforcement Learning based Optimal Policy Autonomous Driving** (06/2019–02/2022) NRF

- *Research on Intelligent Agent-based CPS Security and Reliability* (04/2021–11/2021) TTA
- *Multi-GPU based Automotive HPC Platform Development* (04/2020–10/2020) ETRI
- *Super-Resolution Performance Optimization in Mobile Platforms* (05/2020–08/2020) Samsung SDS
- *Cooperative Deep Reinforcement Learning for Online Game Multi-Agents* (04/2020–08/2020) ETRI
- *Deep Learning Algorithms for mVOC Concentration Analysis* (03/2020–06/2020) Samsung Electronics (C-Lab)
- *Visual Recognition Software Implementation using Deep Learning Tools* (05/2019–11/2019) Hyundai Motors
- *Verification Testbed Implementation for Privacy-Preserving Trust Data Generation* (10/2019–11/2019) ETRI
- *Measurement and Analysis of Multi-Task GPU Scheduling Delays* (05/2019–10/2019) ETRI
- *mmWave High-Speed Networking Platform Design for Next-Generation Convergence Services* (06/2016–05/2019) NRF
- *Probabilistic Decision Making and Econometric Methods for Micro-Grid* (05/2017–04/2019) KEPCO Research Institute
- *GPU Scheduling Performance Analysis under Queueing Delay Considerations* (05/2018–10/2018) ETRI
- *Improving Massive Deep Learning Training via Computation and Communication Acceleration* (04/2018–10/2018) ETRI
- *A Priori Techniques Research for Efficient Multi-Edge Computing* (06/2017–12/2017) Samsung Electronics (Software Center)
- *Feasibility Study of 60 GHz IEEE 802.11ad for Virtual Reality (VR) Platforms* (04/2017–12/2017) IITP
- *Parsing Techniques for Artificial Neural Network (ANN) Data Processing* (09/2017–11/2017) ETRI

University of Southern California, Selected Projects

- *Video Aware Wireless Networks (VAWN) Research Program* Intel Labs, Verizon Wireless, Cisco Systems
- *60 GHz Real-Time Wireless Video Broadcasting* Disney Research Zürich
- *Annenberg Graduate Fellowship Award* University of Southern California

Korea University (KU), Selected Projects

- *Towards Agentic AI-Enabled Edge IoT Systems for Proactive Physical World Sensing and Control* (08/2025–07/2026) PKU-KU Joint Research Grant (Collaboration with Prof. Kaigui Bian at Peking Univ)
- *AI Teaching Assistant Research for LLM-based Large-Scale Education* (10/2024–02/2025) KU University College
- *Autonomous Mobility Control using Quantum Deep Learning* (03/2023–02/2024) KU Insung Research Grant
- *Mobile Access Algorithm Design using Economic Theory and AI* (03/2020–02/2021) KU Future Research Grant

Selected Publications

- **12,186+ Citations** (H-index: 54+, i10-index: 254+), obtained from Google Scholar Profile (as of August 11, 2026)
- **Journals and Magazines:** **177** publications; among them, **132** IEEE/Nature/APS publications

Books

- *Fundamentals of 6G Communications and Networking*, Springer (2024) (Editors: X. Lin, J. Zhang, Y. Liu, J. Kim)

Selected Papers

■ Top-Tier Conference Publications

- [CIKM'26] *Strong Lottery Ticket Hypothesis for Quantum Neural Networks*
ACM CIKM 2026 (H. Kwon, H. K. Lee, J. Kim, S. W. Yoon) (Acceptance Rate: 26.94%)
- [CIKM'26] *Beneath the Noise: Uncovering and Distilling the Encoding Bottleneck of Quantum Models*
ACM CIKM 2026 (S. Lee, C. Kim, S. B. Son, J. Kim, H. P. In) (Acceptance Rate: 30.89%)
- [ICML'26] *Quantum Robust Inner Minimization for Reinforcement Learning with Quadratic Speed-Up in Query Complexity*
ICML 2026 (H. K. Lee, J. Kim, S. W. Yoon) (Acceptance Rate: 26.56%)
- [ACL'26] *Let LLM Tutors Ask First: Proactive LLM-Based Tutoring at Scale in a 1,500-Student Online Classroom*
ACL 2026 (J. Lee, G. Youn, S. Lee, C. Im, J. Kim, C. Yoo)
- [CIKM'25] *Quantum-Amplitude Embedded Adaptation for Parameter-Efficient Fine-Tuning in Large Language Models*
ACM CIKM 2025 (E. J. Roh, J. Kim) (Acceptance Rate: 30.63%)
- [CIKM'25] *Filtered One-Shot Training for Quantum Architecture Search*
ACM CIKM 2025 (S. B. Son, S. Y.-C. Chen, J. Kim, S. Park) (Acceptance Rate: 30.63%)
- [CIKM'25] *LLM-based Interactive Coding Education via Predictive Query Management and Student-Centered Fine-Tuning: Design and Implementation with 1500-Student Class Data*
ACM CIKM 2025 (G. Youn, J. Lee, J. Kim, C. Yoo) (Acceptance Rate: 30.63%)
- [IPDPS'25] *AQUA: Hardware-Agnostic Qubit Allocation for Quantum Multi-Programming*
IEEE IPDPS 2025 (X. Piao, J. Shim, J. Kim, J. Kim) (Acceptance Rate: 24.71%)
- [WiOpt'25] *Stabilized Robust Control for Lightweight Autonomous Aircraft Mobility: A Quantum Reinforcement Learning Approach*
IEEE/IFIP WiOpt 2025 (G. S. Kim, J. Chung, T. Q. Duong, S. Park, J. Kim)
- [NOMS'25] *Joint Multi-Agent Reinforcement Learning and Message-Passing for Distributed Multi-UAV Network Management using Conflict Graphs*
IEEE/IFIP NOMS 2025 (Y. Cho, H. Lee, S. Park, J. Kim)
- [CIKM'24] *Hands-On Introduction to Quantum Machine Learning*
ACM CIKM 2024 (Tutorial) (S. Y.-C. Chen, J. Kim)
- [WiOpt'24] *Advanced Taxiing Path Guidance using Multi-Agent Reinforcement Learning for Air Traffic Management*
IEEE/IFIP WiOpt 2024 (S. Lee, G. S. Kim, S. Park, J. Kim)
- [CIKM'23] *Quantum Split Learning for Privacy-Preserving Information Management*
ACM CIKM 2023 (S. Park, H. Baek, J. Kim) (Acceptance Rate: 27.44%)

- [CIKM'23] *Logarithmic Dimension Reduction for Quantum Neural Networks*
ACM CIKM 2023 (H. Baek, S. Park, J. Kim) (Acceptance Rate: 27.44%)
- [AAAI'23] *Quantum Multi-Agent Meta Reinforcement Learning*
AAAI 2023 (W. Yun, J. Park, J. Kim) (Acceptance Rate: 19.61%)
- [CIKM'22] *Hierarchical Reinforcement Learning using Gaussian Random Trajectory Generation in Autonomous Furniture Assembly*
ACM CIKM 2022 (W. Yun, D. Mohaisen, S. Jung, J.-K. Kim, J. Kim) (Acceptance Rate: 29.64%)
- [WiOpt'22] *Cooperative Video Quality Adaptation for Delay-Sensitive Dynamic Streaming using Adaptive Super-Resolution*
IEEE/IFIP WiOpt 2022 (M. Choi, W. Yun, J. Kim)
- [INFOCOM'22] *Joint Superposition Coding and Training for Federated Learning over Multi-Width Neural Networks*
IEEE INFOCOM 2022 (H. Baek, W. Yun, Y. Kwak, S. Jung, M. Ji, M. Bennis, J. Park, J. Kim) (Acceptance Rate: 19.93%)
- [ICDCS'20] *Understanding the Potential Risks of Sharing Elevation Information on Fitness Applications*
IEEE ICDCS 2020 (Ü. Meteriz, N. F. Yildiran, J. Kim, D. Mohaisen) (Acceptance Rate: 17.98%)
- [IJCAI'19] *Randomized Adversarial Imitation Learning for Autonomous Driving*
IJCAI 2019 (M. Shin, J. Kim) (Acceptance Rate: 17.89%)
- [ICDCS'18] *ShmCaffe: A Distributed Deep Learning Platform with Shared Memory Buffer for HPC Architecture*
IEEE ICDCS 2018 (S. Ahn, J. Kim, E. Lim, W. Choi, A. Mohaisen, S. Kang) (Acceptance Rate: 20.63%)
- [ACM-MM'17] *REQUEST: Seamless Dynamic Adaptive Streaming over HTTP for Multi-Homed Smartphone under Resource Constraints*
ACM Multimedia 2017 (J. Koo, J. Yi, J. Kim, M. Hoque, S. Choi) (Acceptance Rate: 27.63%)
- [MobiSys'10] *Energy-Efficient Rate-Adaptive GPS-based Positioning for Smartphones*
ACM MobiSys 2010 (J. Paek, J. Kim, R. Govindan) (Acceptance Rate: 19.84%)

■ Magazines and Journals

◀ Accepted ▶

- [CIM.accepted] *Rethinking Privacy-Preserving Distributed Learning in Quantum Hilbert Spaces*, *IEEE Computational Intelligence Magazine*, v(n):ppp-ppp (S. B. Son, J. Kim, S. Park)
- [CM.accepted] *Quantum Learning for Autonomous Aircraft Control: A Reinforcement Learning Perspective*, *IEEE Communications Magazine*, v(n):ppp-ppp (G. S. Kim, J. Chung, S. Park, S. Jung, T. Q. Duong, J. Kim)
- [NM.accepted] *Quantum Jump to Connected Worlds: Paradigm Shifts in Mobility and Network Applications via Quantum Neural Networks*, *IEEE Network*, v(n):ppp-ppp (H. Lee, G. S. Kim, E. J. Roh, S. Jung, S. Park, T. Q. Duong, J. Kim)
- [INM.accepted] *Toward Quantum Combinatorial Optimization: From QAOA Fundamentals to Industrial Deployment*, *IEEE Nanotechnology Magazine*, v(n):ppp-ppp (Y. P. Hong, J. Gil, D. Kim, S. Oh, B. S. Kim, J. Kim)
- [TMC.accepted] *Scale-Reconfigurable Multi-Agent Reinforcement Learning for Aerial Networks in Dynamic Environments*, *IEEE Transactions on Mobile Computing*, v(n):ppp-ppp (G. S. Kim, E. J. Roh, S. Jung, S. Park, J. Kim)
- [TMC.accepted] *Efficient Quantum Neural Architecture Search with Filtered Supernet for Mobile Learning Systems*, *IEEE Transactions on Mobile Computing*, v(n):ppp-ppp (S. B. Son, H. Ahn, S. Y.-C. Chen, S. Park, J. Kim)
- [IOTJ.accepted] *USD2C: UAV-Assisted Secure Data Collection and Computation Framework for Internet of Multimedia Things (IoMT) Applications*, *IEEE Internet of Things Journal*, v(n):ppp-ppp (I. Ahmad, J. Kim, S. Shin)
- [AC.accepted] *Physics-Validated Action Masking with LLM-Guided Decomposition for Autonomous Excavation*, *Automation in Construction*, v(n):ppp-ppp (J. Cho, M. Shin, B. Kim, B. Cha, J. Kim, S. Jung)

◀ 2026 ▶

- [TVT'26.10] *Time-to-Collision Aware Autonomous Driving Motion Control under Risky Scenario Considerations*, *IEEE Transactions on Vehicular Technology*, 75(10):ppp-ppp (S. Kim, J. Kim, J. Kim, S. Jung)
- [IOTJ'26.09] *Exponentially Scalable Quantum Reinforcement Learning for Large-Scale Network Scheduling*, *IEEE Internet of Things Journal*, 13(18):ppp-ppp (G. S. Kim, E. J. Roh, S. Park, J. Kim)
- [FGCS'26.09] *GraMA: A Gradient Matrix-Guided Assignment Method for Solving Qubit Mapping Problems*, *Future Generation Computer Systems*, 182:108485 (X. Piao, J. Kim, J.-K. Kim)
- [CM'26.08] *Sustainable 6G Non-Terrestrial Networks: Enabling Net-Zero Operation via Renewable Energy Integration and Intelligent Control*, *IEEE Communications Magazine*, 64(8):110–116 (H. Lee, S. Park, S. Jung, J. Kim)
- [IOTJ'26.08] *Learning-based Resilient Dynamic Routing for Fault-Tolerant Robust LEO Satellite Networks*, *IEEE Internet of Things Journal*, 13(16):37402–37414 (G. S. Kim, C. Im, Y. G. Kim, J. Ha, J. Kim)
- [EAAI'26.08] *A Comprehensive Review on Quantum Deep Neural Networks for Prognostics and Health Management: Fundamentals, Challenges and Opportunities*, *Engineering Applications of Artificial Intelligence*, 177:114991 (M. Jha, S. Chen, C. Kulkarni, J. Kim)
- [TMC'26.07] *Joint Sustainable Control and Quantum Reinforcement Learning for Energy-Efficient Cube-Satellite Networks*, *IEEE Transactions on Mobile Computing*, 25(7):10385–10401 (S. Park, G. S. Kim, Y. Cho, S. Jung, Z. Han, J. Kim)
- [Access'26.07] *Two-Stage MPC-Guided Imitation and Reinforcement Learning for Autonomous Underwater Vehicle Control in Dynamic Obstacle Environments*, *IEEE Access*, 14:99636–99648 (S. Kim, E. J. Roh, I. Song, J. Kwon, S. Kwon, Y. Kim, J. Kim)
- [IOTJ'26.06] *Quantum Federated Gradient Aggregation Using the Parameter-Shift Rule*, *IEEE Internet of Things Journal*, 13(11):25367–25379 (J. Chung, C. Im, S. Park, J. Kim, W. Lee)
- [Nature'26.05] *Fourier Analysis Perspective on Quantum Neural Networks*, *Communications Physics (Nature)*, 9:176 (S. Oh, E. J. Roh, A. Vasilakos, S. Park, J. Kim)
- [TVT'26.04] *Large-Scale Battery-Conscious Collision-Free Path-Finding for Sustainable and Autonomous Multi-AGV Mobility Control*, *IEEE Transactions on Vehicular Technology*, 75(4):5364–5375 (Y. Cho, S. Ahn, S. Park, J. Kim)
- [TAES'26.03] *Quantum Reinforcement Learning for Joint Control, Communication, and Computing in Stabilized Reusable Space Rocket*, *IEEE Transactions on Aerospace and Electronic Systems*, 62:3838–3863 (G. S. Kim, J. Chung, S. Jung, S. Park, J. Kim)

- [IOTJ'26.03] Gateway-Assisted Neural Angular Routing for Hierarchical Satellite Networks, *IEEE Internet of Things Journal*, 13(6):12322–12339 (J. Jang, I.-S. Cho, M. Shin, J. Kim, S. Jung)
- [IOTJ'26.02] Hybrid Large Language Model and Reinforcement Learning for Energy-Efficient Multisatellite Scheduling: Boosting the Performance from Scratch, *IEEE Internet of Things Journal*, 13(3):5379–5392 (H. Ahn, G. S. Kim, I.-S. Cho, S. Jung, J. Kim)
- [PRA'26.02] Understanding Qubit Trajectories in Parametrized Quantum Circuits, *Physical Review A*, 113:022609 (S. Oh, C. Im, S. Park, J. Kim)
- [AC'26.02] Collaborative Learning Architecture for Autonomous Excavator Planning and Execution, *Automation in Construction*, 182:106742 (J. Cho, M. Shin, J. Kim, S. Jung)
- [TAES'26.01] Integrated Control, Communication, and Computing for Mission-Critical Embedded Unmanned Aerial Vehicles, *IEEE Transactions on Aerospace and Electronic Systems*, 62:682–703 (G. S. Kim, S. Park, S. Jung, D. Mohaisen, J. Kim)
- [TMC'26.01] Quantum Multi-Agent Reinforcement Learning for Cooperative Mobile Access in Space-Air-Ground Integrated Networks, *IEEE Transactions on Mobile Computing*, 25(1):1200–1218 (G. S. Kim, Y. Cho, J. Chung, S. Park, S. Jung, Z. Han, J. Kim)
- ◀ 2025 ▶
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- [TS'15.05] Dynamic Two-Stage Beam Training for Energy-Efficient Millimeter-Wave 5G Cellular Systems, *Telecommunication Systems*, 59(1):111–122 (J. Kim, S.-N. Hong)
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- [CL'14.03] Fast and Low-Power Link Setup for IEEE 802.15.3c Multi-Gigabit/s Wireless Sensor Networks, *IEEE Communications Letters*, 18(3):455–458 (J. Kim, A. Mohaisen, J.-K. Kim)
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■ IEEE ComSoc, VTS, and Computer Society Conference Contributions

- 50 papers at INFOCOM, ICDCS, GLOBECOM, ICC, WCNC, CCNC, PIMRC, VTC, APWCS, IEEE Quantum Week
- XX papers at other ComSoc, VTS, and Computer Society conferences

Academic Activities (Selected: **Quantum AI and Quantum Computing**)

Invited Talks at Universities (Selected)

- Quantum Neural Networks: Algorithms and Applications
Keio University (Tokyo, Japan, 12/2025)
- Quantum Reinforcement Learning: From Foundations to Emerging Applications
Ulsan National Institute of Science and Technology (UNIST) (Ulsan, Korea, 12/2025)
- Quantum Reinforcement Learning: Algorithms and Applications
Indiana University, Bloomington (Bloomington, Indiana, USA, 10/2025)
- Quantum Neural Networks: Theory, SW, and Applications
Gwangju Institute of Science and Technology (GIST) (Gwangju, Korea, 05/2025)
- Quantum Neural Networks: Theory, SW, and Applications
Seoul National University Hospital (Seoul, Korea, 03/2025)
- Quantum Neural Networks: Theory, SW, and Applications
Sungkyunkwan University (Suwon, Korea, 01/2025)
- Quantum Neural Networks: Theory, SW, and Applications
Yonsei University (Seoul, Korea, 09/2023)

Journal Special Issues (Guest Editorials)

- Quantum Technologies for Communication Systems (2025), *IEEE/KICS Journal of Communications and Networks*
- Advances in Quantum Networks: Enabling Computing, Security, and AI Innovations (2025), *Discover Networks (Springer)*
- Quantum Algorithms, Optimization, and A.I. (2025), *The Journal of Supercomputing (Springer)*

Organizing Committee Contributions for International Conferences, Symposiums, and Workshops

- Quantum Neural and Reinforcement Learning for Adaptive Sequential Intelligence (ID 47)
IEEE QAI 2026 (Nottingham, United Kingdom, 12/2026)
- Workshop on Quantum Computing and Reinforcement Learning (2025)
IEEE Quantum Week (QCE) 2026 (Toronto, Ontario, Canada, 09/2026)
- Quantum Technologies for Earth Observation (2026)
IEEE IGARSS 2026 (Washington DC, USA, 08/2026)
- Special Session: Quantum Machine Learning Algorithms and Applications (2026)
IEEE WCCI 2026 (Maastricht, The Netherlands, 06/2026)
- Workshop on Quantum Machine Learning in Signal Processing and Artificial Intelligence (2026)
IEEE ICASSP 2026 (Barcelona, Spain, 04/2026)
- Workshop: Quantum Computing - Artificial Intelligence (QC-AI) for Industry Applications and Scientific Discovery (2026)
IEEE QCNC 2026 (Kobe, Japan, 04/2026)
- Quantum Computing for Classical Computational Challenges in HPC (2026)
IEEE QCNC 2026 (Kobe, Japan, 04/2026)
- Workshop on Quantum Machine Learning (2026)
IEEE ICOIN 2026 (Hanoi, Vietnam, 01/2026)
- Workshop on Quantum Computing for Communications and Learning (2025)
IEEE GLOBECOM 2025 (Taipei, Taiwan, 12/2025)
- Workshop on Quantum Computing and Reinforcement Learning (2025)
IEEE Quantum Week (QCE) 2025 (Albuquerque, New Mexico, USA, 09/2025)
- Workshop on Quantum Machine Learning in Signal Processing and Artificial Intelligence (2025)
IEEE ICASSP 2025 (Hyderabad, India, 04/2025)
- Workshop on Quantum Computing and Reinforcement Learning (2024)
IEEE Quantum Week (QCE) 2024 (Montreal, Quebec, Canada, 09/2024)
- Workshop on Quantum Algorithms, Optimization, and Artificial Intelligence (2024)
IJCAI 2024 (Jeju, Korea, 08/2024)

Program Committee Contributions for International Conferences, Symposiums, and Workshops

- Workshop on Secure and Trustworthy Quantum Machine Learning (2026)
NeurIPS 2026 (Atlanta, Georgia, USA, 12/2026)
- Workshop on Quantum Machine Learning (2026)
IEEE ICOIN 2026 (Hanoi, Vietnam, 01/2026)
- Workshop on Quantum Computing for Communications and Learning (2025)
IEEE GLOBECOM 2025 (Taipei, Taiwan, 12/2025)
- Workshop on Quantum Data and Machine Learning: Systems, Theory and Hardware (2025)
IEEE ICDE 2025 (Hong Kong, China, 05/2025)
- Workshop on Quantum Computing and Artificial Intelligence (2024)
AAAI 2024 (Philadelphia, Pennsylvania, USA, 02/2024)

International Tutorials and Speeches

- Hands-On Quantum AI for Sequential Decision-Making: Variational Quantum Models, Quantum Reinforcement Learning, and Self-Programming Quantum Circuits (ID 46)
IEEE QAI 2026 Tutorial (Nottingham, United Kingdom, 12/2026)
- Introduction to Distributed Quantum Artificial Intelligence: Neural Networks, Reinforcement Learning, and Multi-Agent Systems
IEEE WCCI 2026 Tutorial (Maastricht, The Netherlands, 06/2026)

- Quantum-Enhanced Learning for Future Communication Networks
IEEE GLOBECOM 2025 Tutorial (Taipei, Taiwan, 12/2025)
- Quantum Reinforcement Learning: From Foundations to Emerging Applications
IEEE QAI 2025 Tutorial (Naples, Italy, 11/2025)
- Quantum Machine Learning: Algorithms and ICT Convergence Applications
IEEE ICTC 2025 Invited Talk (Jeju, Korea, 10/2025)
- Hands-On Introduction to Quantum Machine Learning
IEEE QCE 2025 Tutorial (Albuquerque, New Mexico, USA, 09/2025)
- Quantum Reinforcement Learning for Computer Networks and Autonomous Mobility: Modern Trends and Future Directions
IEEE VTS APWCS 2025 Invited Speech (Tokyo, Japan, 08/2025)
- Quantum Reinforcement Learning: Algorithms and Applications
IEEE Future Networks Webinar (Future Networks AI/ML and QIT Working Groups) (Webinar, 08/2025)
- Introduction to Quantum Machine Intelligence
IEEE SSCI 2025 Tutorial (Trondheim, Norway, 03/2025)
- Hands-On Introduction to Quantum Machine Learning
ACM CIKM 2024 Tutorial (Boise, Idaho, USA, 10/2024)
- Quantum Reinforcement Learning: Concepts, Models, and Applications
ICIT 2024 Keynote Speech (Danang, Vietnam, 09/2024)
- A Paradigm Shift in Future Networks with Quantum Deep Learning
IEEE ICUFN 2022 Tutorial (Barcelona, Spain, 07/2022)

Academic Activities and Research Supervision

Research Supervision

■ Ph.D. Alumni

- Soohyun Park (03/2019–08/2023 (PhD), 09/2023–02/2024 (Postdoc)), **Sookmyung Women's University (Professor)**
- Hankyul Baek (03/2021–02/2024 (PhD), 03/2024–03/2025 (Postdoc)), ETRI AI Safety Institute (Researcher)
- Hyunsoo Lee (03/2021–02/2026 (PhD)), LG Electronics (Researcher)
- Gyu Seon Kim (03/2023–02/2026 (PhD), 03/2026–04/2026 (Postdoc)), RealtimeVisual (Researcher, *Military Service Exception*)

■ M.S. Alumni

- Kyeongseon Kim (09/2017–08/2019), KAIST (Ph.D. Student in Computer Science)
- Dohyun Kwon (03/2018–02/2020), Hyundai Marine & Fire Insurance (Researcher)
- Dohyun Kim (03/2018–02/2020), Naver Webtoon (Researcher)
- MyungJae Shin (03/2018–02/2020), Naver (Researcher)
- Jaeho Choi (03/2019–02/2021), Korea Meteorological Administration (Researcher, *Military Service Exception*)
- Yoo Jeong (Anna) Ha (03/2021–02/2023), The University of Chicago (Ph.D. Student in Computer Science)
- Jaehyun Chung (09/2023–08/2025), Hanwha Aerospace (Researcher)
- Chaemoon Im (03/2024–02/2026), Korea University Net-Zero CAFE Research Center (Researcher)
- Yeryeong Cho (03/2024–02/2026), Korea University Net-Zero CAFE Research Center (Researcher)

■ Former Research Professors and Postdoctoral Scholars

- Minseok Choi (09/2018–02/2019, jointly with Prof. Andreas F. Molisch (USC)), **Kyung Hee University (Professor)**
- Soyi Jung (03/2021–08/2021, jointly with Prof. Marco Levorato (UC-Irvine)), **Ajou University (Professor)**
- Ju-Hyung Lee (03/2022–02/2023, jointly with Prof. Andreas F. Molisch (USC)), Nokia USA (Researcher)

■ Former Full-Time Researchers

- Hyomin Ahn (09/2020–01/2021), UCLA (Ph.D. Student in Electrical and Computer Engineering)
- Rhoan Lee, LG CNS (Researcher) (M.S. in Statistics, Columbia University in the City of New York)
- Hyunhee Cho, KAIST (Ph.D. Student in Electrical Engineering)
- Daeyeun Kim (07/2025–07/2026), The University of Manchester (Ph.D. Student in Mechanical Engineering)
- Donggun Lee (04/2026–08/2026), The University of Michigan (B.S. Student in Data Science)

IEEE Activities (Membership, Editorial Boards, and Services)

- Senior Member (2018–), Member (2006–2017)
- Associate Editor (2025–), **ACM Computing Surveys**
- Associate Editor (2025–), **IEEE Communications Surveys and Tutorials**
- Editor (2023–), **IEEE Internet of Things Journal**
- Associate Editor (2020–), **IEEE Transactions on Vehicular Technology**
- Guest Editor, **Journal of Communications and Networks** (S.I. on Quantum Technologies for Communication Systems)
- Guest Editor, **IEEE Communications Standards Magazine** (S.I. on Recent and Future Evolution of Wi-Fi)

References

- Prof. Andreas F. Molisch (*Fellow of the IEEE*), Ph.D. Research and Dissertation Advisor
– Professor of Electrical and Computer Engineering at the University of Southern California (Los Angeles, CA, USA)
– E-mail: molisch@usc.edu